SCIENCE EXPLAINERS

Science explainers present the driving ideas and progress from NextGEMS scientists in plain language. The Explainers aim to show how scientists think and their way of working and cover a wide range of topics relevant to any audience interested in the latest breakthroughs related to strom-resolving Earth Systems Models (SR-ESM).

OCEAN MODELS IN NEXTGEMS:

TACKLING THE BIASES OF TROPICAL MIXED LAYER DEPTH

Author: Markus Jochum and Marta Mrozowska
Affiliation: University of Copenhagen - Niels Bohr Institute
Department: Physics of Ice, Climate and Earth
Expertise: Ocean modelling
NextGEMS theme: Storms & Ocean

WHAT

It is very important to understand ocean-atmosphere feedbacks, especially in the tropics, where slight changes in sea surface temperature cause considerable differences in atmospheric dynamics.

The **ocean boundary layer** is the water mass laying on top of the stratified ocean interior and is considered to be well mixed, ie. constant in density, temperature and salinity along the depth axis. The thickness of this layer, also called the mixed layer, determines the temperature and salinity at the sea surface. Turbulent mixing occurs at sub-grid scales, and therefore needs to be parameterized in the climate models.

"THE BIAS IN THE MIXED LAYER DEPTH IS ONE OF THE KEY CHALLENGES IN TODAY'S CLIMATE MODEL DEVELOPMENT."

WHAT ARE WE WORKING ON?

Near-inertial waves (NIWs) are resonant modes of the ocean mixed layer, in which water parcels describe circular motions at inertial periods. Induced by wind bursts they increase current shear and hence mixing. We think that near-inertial signal might be missing from the current parameterization of the mixed layer, and that including it in the theoretical framework might improve the accuracy of ocean simulations

THIS IS A RELATIVELY UNEXPLORED IDEA. OUR EXPECTATION THAT NIWS ARE IMPORTANT FOR MIXING COMES FROM THE PUBLICATION BY JOCHUM, ET AL. (2013) [1], WHERE ENHANCING THE NIW MIXING SIGNAL RESULTS IN REDUCTION OF TROPICAL SST BIASES IN THE COMMUNITY EARTH SYSTEM MODEL.

WHAT ARE THE CHALLENGES WE FACE?

Mixed layer depth observations are scarce, and high temporal frequency ocean current observations (needed to catch the near-inertial signal) can be difficult to find. Therefore, validation of the model results is a big challenge in this project. The potential opportunity is significantly improving how the ocean-atmosphere feedbacks are represented in general circulation models, which is beneficial to many research areas in the field.

EXPECTED CONTRIBUTION

Improving the tropical ocean-atmosphere feedbacks could in theory improve the predictability of tropical climate. This would mean more reliable predictions for future climate for policymakers, companies and infrastructure.

REFERENCES

Jochum, M., Briegleb, B. P., Danabasoglu, G., Large, W. G., Norton, N. J., Jayne, S. R., ... Bryan, F. O. (2013). The impact of oceanic near-inertial waves on climate. *Journal Of Climate*, 26, 2833-2844. doi:10.1175/JCLI-D-12-00181.1

next GEMS

NEXTGEMS IS FUNDED THROUGH THE EUROPEAN UNION'S HORIZON 2020 RESEARCH AND INNOVATION PROGRAMME UNDER THE GRANT AGREEMENT NO - 101003470